Session 14 · Tuesday 13 October 2026 · Not a practice exam. Five things this course asked you to watch rather than do, each worked once and then handed to you.

Five motions, in the shape they will be asked

There is nothing new here. These are five moves you have seen done — on a slide, on the board, in a worked answer — and have not often done yourself with a pencil. Each one is worked once below, then handed to you on a different circuit.

None of these circuits is on Thursday's paper. The point is the motion, not the example.

Write in the boxes. The box is a guide to how long the answer should be.

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1 — Follow a disturbance around a loop

A loop is a clock only if a disturbance comes back **flipped** and **not shrunk**. Nudge one gene up and walk it round: each step either flips the sign or keeps it, and each step shrinks or amplifies. Come back unflipped and the disturbance reinforces what started it — the circuit latches into a state. Come back shrunk and it dies — the circuit settles.

Worked. Three genes, mixed signs: $X \dashv Y$, $Y \dashv Z$, $Z \longrightarrow X$. Nudge X up. Y goes down. Z goes up. Z **activates** X, so X goes up — **unflipped**. Two flips, an even number, so the disturbance reinforces what began it. This loop cannot be a clock at any gain; it latches.

▶ **Now you.** A **four-gene loop with mixed signs**:

$$X \dashv Y, \qquad Y \longrightarrow Z, \qquad Z \dashv W, \qquad W \dashv X$$

Walk a nudge in X once around and say what comes back.

Can this loop be a clock? If not, say which half of the test it fails and what it does instead. If it can, say what still has to be true.

Why does that step follow? — what is it about the **number** of sign flips, rather than their strength, that settles this?

▶ **And one more — change exactly one edge.** In the same four-gene loop, replace $Z \dashv W$ with $Z \longrightarrow W$ and leave everything else alone. Clock or not? Say what changed and what did not.

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2 — Read the signs of a Jacobian off the wiring

You have computed eigenvalues from numbers all term. This is the other direction: what the **pattern of signs** is, and where each sign comes from — no differentiation.

The rule is short. For $\dot{x}_i = (\text{production}) - x_i$:

- The **diagonal** $\partial \dot{x}_i / \partial x_i$ carries that species' own removal, so it is **negative** — unless the species also acts on itself.
- An **off-diagonal** $\partial \dot{x}_i / \partial x_j$ carries only how j changes i's production. If j **represses** i, raising j lowers the production, so the entry is **negative**. If j **activates** i, the entry is **positive**.

Worked. An **activator–repressor** pair: X activates Y, Y represses X, $\dot{x} = \alpha/(1+y^{\,n}) - x$ and $\dot{y} = \beta x^{\,n}/(1+x^{\,n}) - y$:

$$J = \begin{pmatrix} - & - \\ + & - \end{pmatrix}$$

Both diagonals negative — each protein is removed, and neither acts on itself. Top-right negative: Y represses X, so raising y lowers X's production. Bottom-left **positive**: X activates Y.

▶ **Now you.** A **three-gene cascade**: X activates Y, Y represses Z, and nothing feeds back.

$$\dot{x} = \beta_x - x, \qquad \dot{y} = \frac{\beta\, x^{\,n}}{1 + x^{\,n}} - y, \qquad \dot{z} = \frac{\alpha}{1 + y^{\,n}} - z$$

Fill in the **nine** signs, and say where each comes from. Several are **zero** — say why, because that is the part that is new.

$$J = \begin{pmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{pmatrix}$$

Why does that step follow? — a cascade's Jacobian is **triangular**. Say in one line what that tells you about its eigenvalues without computing any of them.

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3 — Draw a time course

You have watched these on the screen. Drawing one is a different skill, and it is quicker than it looks: what matters is the **shape** and the **relative timing**, not numbers.

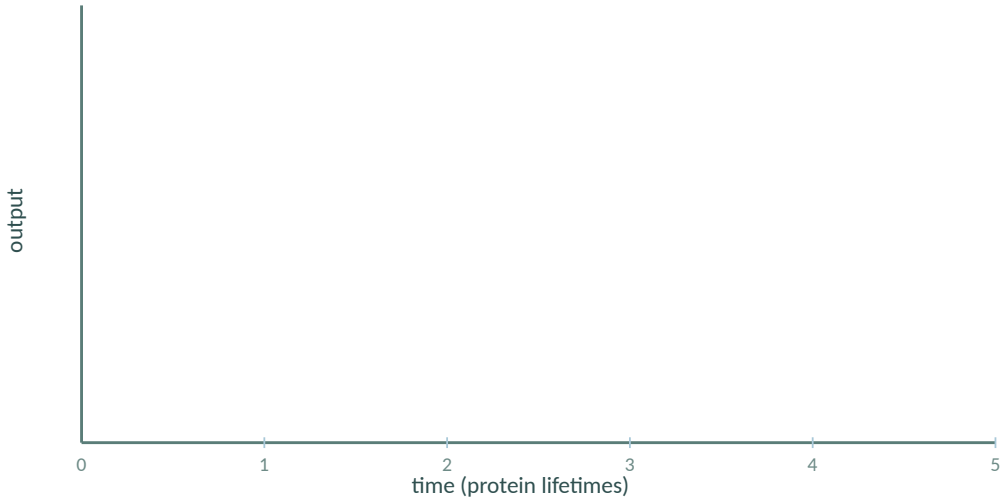
Three things make a curve: where it starts, how fast it leaves, and where it ends. Get those three right and the curve between them is just a smooth line.

Worked. A gene switched on at $t = 0$ with constant production and first-order removal starts at zero, rises fastest at the start, and flattens onto β/γ. Its half-way time is $\ln 2/\gamma$ — one half-life — and it is **within about 6% of its final level after four half-lives**.

▶ **Now you.** On the axes below, draw **two** curves for a gene switched on at $t = 0$:

- **(a)** an unregulated gene, rising to its steady level;
- **(b)** the **same gene with negative autoregulation added** and nothing else touched — same promoter, same removal rate.

Label which is which, and mark each curve's **half-way time** on the time axis.



Why does that step follow? — of the three things that make a curve (start, initial slope, end), say which ones the feedback changed and which it left alone. There is more than one.

4 — Say which state a circuit ends up in

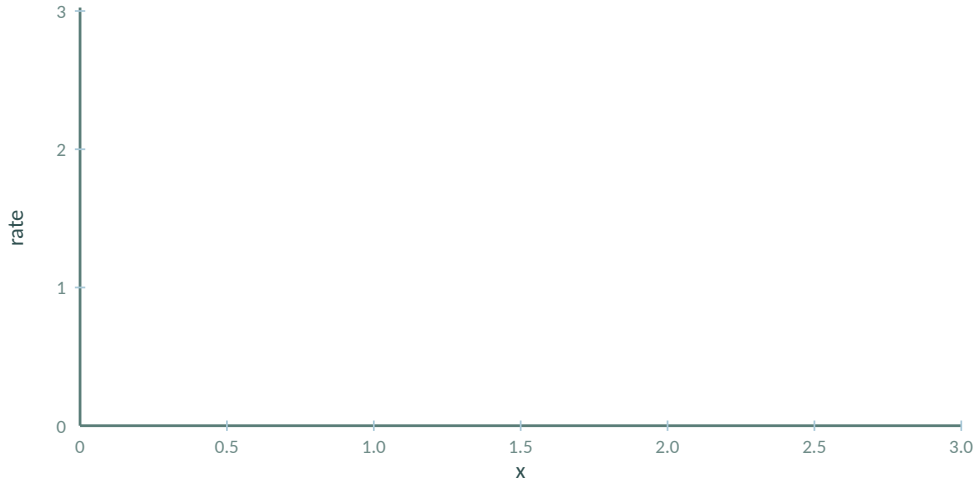
A circuit with two stable states has a **divide** between them, and it always runs *into* an unstable point. Start on one side of it and the flow carries you to one state; start on the other and you reach the other. Nothing else decides it — not how large a disturbance was, only **which side it left you on**. With one variable the divide is a single **number** on the x-axis; with two it is a **curve** in the plane, and that difference is the whole of this item.

Worked — in class. Two genes that **activate each other**, $\alpha = 3$, $n = 2$. Its two states are *both off* and *both on*, with an unstable point between them at $x = y = 0.382$. A pulse raises X **alone**, so the cell is left somewhere on the x-axis with Y still at zero — and the dividing line crosses that axis at $x = 0.711$, nowhere near 0.382. A pulse to $x = 0.5$ is well past the middle point and **still falls back off**. The rule is *which side of the curve*, never *how far along one axis*.

▶ **Now you — a circuit with only one variable.** A single gene **activates its own promoter**, with a small constant basal rate and the usual first-order removal:

$$\dot{x} \;=\; \beta \;+\; \frac{\alpha\, x^{\,n}}{1 + x^{\,n}} \;-\; x, \qquad \beta = 0.2, \quad \alpha = 2, \quad n = 3$$

(a) Sketch the **production** term and the **removal** term on the same axes against x, and mark every point where they cross.



(b) There are three crossings. Which are the **states**, which is the **divide**, and how does your sketch tell you — without computing a single derivative?

(c) The cell sits at the lower state. Give the **number** a brief pulse has to push x past for the cell to switch on and stay on.

Why does that step follow? — here the divide is a **number**; in the two-gene case on the slides it was a **curve**, and reading it as a number on one axis gave the wrong answer. Say in one line what it is about having two variables that makes the difference.

5 — Read a trace off the wiring and the timing

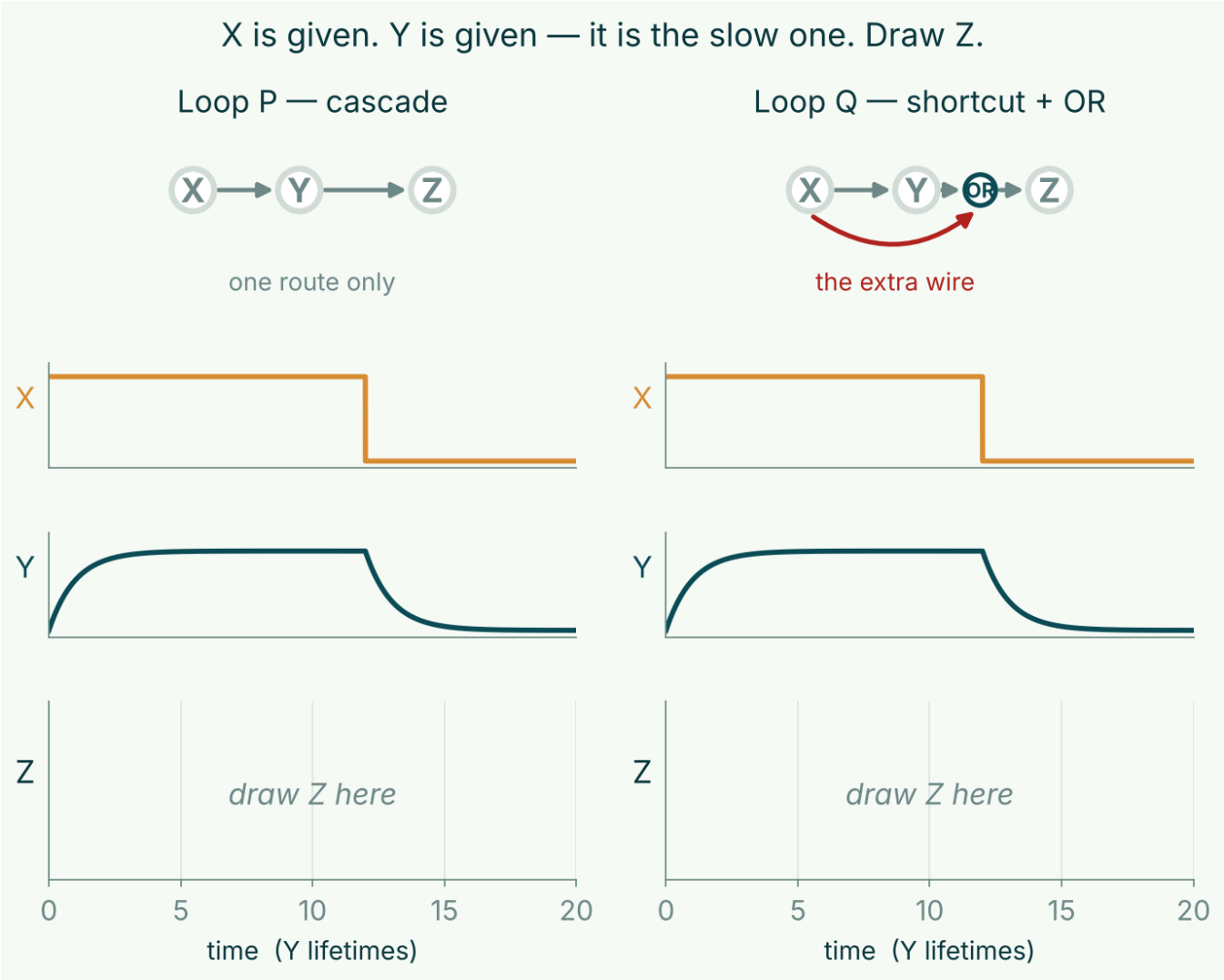
Two circuits take the same input X, which turns **on** at $t = 0$ and **off** twelve lifetimes later. In both, Z follows its own promoter quickly; the intermediate Y is the slow one and sets the clock.

Loop P — a cascade. $X \longrightarrow Y \longrightarrow Z$. Nothing else.

Loop Q — a shortcut and an OR gate. $X \longrightarrow Y \longrightarrow Z$ **and** $X \longrightarrow Z$, with Z switched on by **either** input — X **OR** Y.

Worked, for Loop P. When X arrives, Z cannot move until Y has built up, and Y is slow — so the rise is **late**. When X leaves, Z cannot fall until Y has decayed away, and Y is still slow — so the fall is **late too**. A cascade is late at **both** ends, by roughly the same amount. The delay is symmetric.

▶ **Now you.** Each column below carries its own wiring, its input X, and its slow arm Y. Only the Z panels are empty. Draw Loop P's Z in the left column (you have just been told its shape), then **Loop Q's** in the right.



Then answer, in one line each:

(a) Which end — switching on, or switching off — does the shortcut change, and which does it leave alone?

(b) Why does the OR gate do that? Say it in terms of what Z is waiting for at each end.

(c) A circuit whose delay is different in the two directions is called **sign-sensitive**. If the gate were **AND** instead of OR, which end would be delayed? Do not guess — reason it from what each gate needs.

If you only take one thing from each

1. **Count the flips before you reason about strength.** An even number of inversions cannot be rescued by any amount of gain.
2. **The signs of a Jacobian come from the wiring**, not from differentiating. Own removal on the diagonal; who acts on whom, and in which direction, off it.
3. **A curve is a start, a slope and an end.** Draw those and join them.
4. **Which side, not how far.** A circuit with two states has a line between them, and where a disturbance *leaves* you decides where you end up.
5. **A shortcut changes one end, not both.** Which end depends on the gate — work it out from what the output is waiting for, never from a remembered rule.